

Module 7: Spring REST using Spring Boot 3

1. Create a Spring Web Project using Maven

The Project Directory is as follows:

```
spring-learn
├── .mvn
├── src
│   ├── main
│   │   ├── java
│   │   │   ├── com
│   │   │   │   ├── cognizant
│   │   │   │   │   ├── spring_learn
│   │   │   │   │   │   └── SpringLearnApplication.java
│   │   └── resources
│   └── test
├── target
├── .gitattributes
├── .gitignore
├── HELP.md
├── mvnw
├── mvnw.cmd
└── pom.xml
```

Codes:

SpringLearnApplication.java

```
package com.cognizant.spring_learn;
import org.slf4j.Logger;
import org.slf4j.LoggerFactory;
import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;
@SpringBootApplication
public class SpringLearnApplication {
    private static final Logger LOGGER =
        LoggerFactory.getLogger(SpringLearnApplication.class);
    public static void main(String[] args) {
        SpringApplication.run(SpringLearnApplication.class, args);
        LOGGER.info("Inside main");
    }
}
```

Output:

```
2026-07-26T15:23:41.343+05:30 INFO 1532 --- [spring-learn] [ restartedMain] o.s.boot.tomcat.TomcatWebServer : Tomcat started on port 8083 (http) with context path '/'
2026-07-26T15:23:41.356+05:30 INFO 1532 --- [spring-learn] [ restartedMain] c.c.spring_learn.SpringLearnApplication : Started SpringLearnApplication in 3.437 seconds (process running for 4.421)
2026-07-26T15:23:41.365+05:30 INFO 1532 --- [spring-learn] [ restartedMain] c.c.spring_learn.SpringLearnApplication : Inside main
2026-07-26T15:25:03.257+05:30 INFO 1532 --- [spring-learn] [nio-8083-exec-1] o.a.c.c.C.[tomcat].[localhost].[/] : Initializing Spring DispatcherServlet 'dispatcherServlet'
2026-07-26T15:25:03.258+05:30 INFO 1532 --- [spring-learn] [nio-8083-exec-1] o.s.web.servlet.DispatcherServlet : Initializing Servlet 'dispatcherServlet'
2026-07-26T15:25:03.261+05:30 INFO 1532 --- [spring-learn] [nio-8083-exec-1] o.s.web.servlet.DispatcherServlet : Completed initialization in 2 ms
```

2. Spring Core- Load Country from Spring Configuration XML

The Project Directory is as follows:

```
spring-learn
├── .mvn
├── src
│   ├── main
│   │   ├── java
│   │   │   ├── com
│   │   │   │   ├── cognizant
│   │   │   │   │   ├── spring_learn
│   │   │   │   │   │   ├── Country.java
│   │   │   │   │   │   └── SpringLearnApplication.java
│   │   └── resources
│   │       ├── static
│   │       ├── templates
│   │       ├── application.properties
│   │       └── country.xml
│   └── test
│       ├── java
│       │   ├── com
│       │   │   ├── cognizant
│       │   │   │   ├── spring_learn
│       │   │   │   │   └── SpringLearnApplicationTests.java
├── target
├── .gitattributes
├── .gitignore
├── HELP.md
├── mvnw
├── mvnw.cmd
└── pom.xml
```

Codes:

Country.java

```
package com.cognizant.spring_learn;
import org.slf4j.Logger;
import org.slf4j.LoggerFactory;
public class Country {
    private static final Logger LOGGER =
```

```

        LoggerFactory.getLogger(Country.class);
private String code;
private String name;
public Country() {
    LOGGER.debug("Inside Country Constructor.");
}
public String getCode() {
    LOGGER.debug("Inside getCode()");
    return code;
}
public void setCode(String code) {
    LOGGER.debug("Inside setCode()");
    this.code = code;
}
public String getName() {
    LOGGER.debug("Inside getName()");
    return name;
}
public void setName(String name) {
    LOGGER.debug("Inside setName()");
    this.name = name;
}
@Override
public String toString() {
    return "Country [code=" + code + ", name=" + name + "]";
}
}

```

SpringLearnApplication.java

```

package com.cognizant.spring_learn;
import org.slf4j.Logger;
import org.slf4j.LoggerFactory;
import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;
import org.springframework.context.ApplicationContext;
import org.springframework.context.support.ClassPathXmlApplicationContext;
@SpringBootApplication
public class SpringLearnApplication {
    private static final Logger LOGGER =
        LoggerFactory.getLogger(SpringLearnApplication.class);
    public static void main(String[] args) {
        SpringApplication.run(SpringLearnApplication.class, args);
        LOGGER.info("Inside main");
        displayCountry();
    }
    public static void displayCountry() {
        LOGGER.info("Start");
        ApplicationContext context =
            new ClassPathXmlApplicationContext("country.xml");
    }
}

```

```

        Country country =
            context.getBean("country", Country.class);
        LOGGER.debug("Country : {}", country);
        LOGGER.info("End");
        ((ClassPathXmlApplicationContext) context).close();
    }
}

```

application.properties

```

logging.level.org.springframework=INFO
logging.level.com.cognizant=DEBUG
server.port=8083

```

country.xml

```

<?xml version="1.0" encoding="UTF-8"?>
<beans xmlns="http://www.springframework.org/schema/beans"
       xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
       xsi:schemaLocation="
           http://www.springframework.org/schema/beans
           https://www.springframework.org/schema/beans/spring-beans.xsd">
    <bean id="country"
          class="com.cognizant.spring_learn.Country">
        <property name="code" value="IN"/>
        <property name="name" value="India"/>
    </bean>
</beans>

```

SpringLearnApplicationTests.java

```

package com.cognizant.spring_learn;
import org.junit.jupiter.api.Test;
import org.springframework.boot.test.context.SpringBootTest;
@SpringBootTest
class SpringLearnApplicationTests {

    @Test
    void contextLoads() {
    }

}

```

Output:

```

PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS
2026-07-26T16:18:30.085+05:30 INFO 12676 --- [ restartedMain] o.s.boot.tomcat.TomcatWebServer : Tomcat initialized with port 8083 (http)
2026-07-26T16:18:30.108+05:30 INFO 12676 --- [ restartedMain] o.apache.catalina.core.StandardService : Starting service [Tomcat]
2026-07-26T16:18:30.108+05:30 INFO 12676 --- [ restartedMain] o.apache.catalina.core.StandardEngine : Starting Servlet engine: [Apache Tomcat/11.0.22]
2026-07-26T16:18:30.163+05:30 INFO 12676 --- [ restartedMain] b.w.c.s.WebApplicationContextInitializer : Root WebApplicationContext: initialization completed in 1075 ms
2026-07-26T16:18:30.628+05:30 INFO 12676 --- [ restartedMain] o.s.boot.tomcat.TomcatWebServer : Tomcat started on port 8083 (http) with context path '/'
2026-07-26T16:18:30.628+05:30 INFO 12676 --- [ restartedMain] c.c.spring_learn.SpringLearnApplication : Started SpringLearnApplication in 2.299 seconds (process running for 2.939)
2026-07-26T16:18:30.633+05:30 INFO 12676 --- [ restartedMain] c.c.spring_learn.SpringLearnApplication : Inside main
2026-07-26T16:18:30.633+05:30 INFO 12676 --- [ restartedMain] c.c.spring_learn.SpringLearnApplication : Start
2026-07-26T16:18:30.767+05:30 DEBUG 12676 --- [ restartedMain] com.cognizant.spring_learn.Country : Inside Country Constructor.
2026-07-26T16:18:30.783+05:30 DEBUG 12676 --- [ restartedMain] com.cognizant.spring_learn.Country : Inside setCode()
2026-07-26T16:18:30.783+05:30 DEBUG 12676 --- [ restartedMain] com.cognizant.spring_learn.Country : Inside setName()
2026-07-26T16:18:30.783+05:30 DEBUG 12676 --- [ restartedMain] c.c.spring_learn.SpringLearnApplication : Country : Country [code=IN, name=India]
2026-07-26T16:18:30.783+05:30 INFO 12676 --- [ restartedMain] c.c.spring_learn.SpringLearnApplication : End

```

3. Hello World RESTful Web Service

The Project Directory is as follows:

```
spring-learn
├── .mvn
├── src
│   ├── main
│   │   ├── java
│   │   │   ├── com
│   │   │   │   ├── cognizant
│   │   │   │   │   ├── spring_learn
│   │   │   │   │   │   ├── controller
│   │   │   │   │   │   │   ├── HelloController.java
│   │   │   │   │   │   │   ├── Country.java
│   │   │   │   │   │   │   └── SpringLearnApplication.java
│   │   │   │   │   └── spring_learn
│   │   │   │   └── cognizant
│   │   │   └── com
│   │   └── resources
│   │       ├── static
│   │       ├── templates
│   │       ├── application.properties
│   │       └── country.xml
│   └── test
│       ├── java
│       │   ├── com
│       │   │   ├── cognizant
│       │   │   │   ├── spring_learn
│       │   │   │   └── SpringLearnApplicationTests.java
│       └── cognizant
├── target
├── .gitattributes
├── .gitignore
├── HELP.md
├── mvnw
├── mvnw.cmd
└── pom.xml
```

Codes:

HelloController.java

```
package com.cognizant.spring_learn.controller;
import org.slf4j.Logger;
import org.slf4j.LoggerFactory;
import org.springframework.web.bind.annotation.GetMapping;
import org.springframework.web.bind.annotation.RestController;
@RestController
public class HelloController {
    private static final Logger LOGGER =
        LoggerFactory.getLogger(HelloController.class);
```

```

    @GetMapping("/hello")
    public String sayHello() {
        LOGGER.info("Start");
        LOGGER.info("End");
        return "Hello World!!";
    }
}

```

Country.java

```

package com.cognizant.spring_learn;
import org.slf4j.Logger;
import org.slf4j.LoggerFactory;
public class Country {
    private static final Logger LOGGER =
        LoggerFactory.getLogger(Country.class);
    private String code;
    private String name;
    public Country() {
        LOGGER.debug("Inside Country Constructor.");
    }
    public String getCode() {
        LOGGER.debug("Inside getCode()");
        return code;
    }
    public void setCode(String code) {
        LOGGER.debug("Inside setCode()");
        this.code = code;
    }
    public String getName() {
        LOGGER.debug("Inside getName()");
        return name;
    }
    public void setName(String name) {
        LOGGER.debug("Inside setName()");
        this.name = name;
    }
    @Override
    public String toString() {
        return "Country [code=" + code + ", name=" + name + "]";
    }
}

```

SpringLearnApplication.java

```

package com.cognizant.spring_learn;
import org.slf4j.Logger;
import org.slf4j.LoggerFactory;
import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;
import org.springframework.context.ApplicationContext;

```

```

import org.springframework.context.support.ClassPathXmlApplicationContext;
@SpringBootApplication
public class SpringLearnApplication {
    private static final Logger LOGGER =
        LoggerFactory.getLogger(SpringLearnApplication.class);
    public static void main(String[] args) {
        SpringApplication.run(SpringLearnApplication.class, args);
        LOGGER.info("Inside main");
        displayCountry();
    }
    public static void displayCountry() {
        LOGGER.info("Start");
        ApplicationContext context =
            new ClassPathXmlApplicationContext("country.xml");
        Country country =
            context.getBean("country", Country.class);
        LOGGER.debug("Country : {}", country);
        LOGGER.info("End");
        ((ClassPathXmlApplicationContext) context).close();
    }
}

```

application.properties

```

logging.level.org.springframework=INFO
logging.level.com.cognizant=DEBUG
server.port=8083

```

country.xml

```

<?xml version="1.0" encoding="UTF-8"?>
<beans xmlns="http://www.springframework.org/schema/beans"
    xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
    xsi:schemaLocation="
        http://www.springframework.org/schema/beans
        https://www.springframework.org/schema/beans/spring-beans.xsd">
    <bean id="country"
        class="com.cognizant.spring_learn.Country">
        <property name="code" value="IN"/>
        <property name="name" value="India"/>
    </bean>
</beans>

```

SpringLearnApplicationTests.java

```

package com.cognizant.spring_learn;
import org.junit.jupiter.api.Test;
import org.springframework.boot.test.context.SpringBootTest;
@SpringBootTest
class SpringLearnApplicationTests {
    @Test
    void contextLoads() {
    }
}

```

```
}
```

pom.xml

```
<?xml version="1.0" encoding="UTF-8"?>
<project xmlns="http://maven.apache.org/POM/4.0.0"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
    xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
https://maven.apache.org/xsd/maven-4.0.0.xsd">
  <modelVersion>4.0.0</modelVersion>
  <parent>
    <groupId>org.springframework.boot</groupId>
    <artifactId>spring-boot-starter-parent</artifactId>
    <version>4.1.0</version>
    <relativePath/> <!-- lookup parent from repository -->
  </parent>
  <groupId>com.cognizant</groupId>
  <artifactId>spring-learn</artifactId>
  <version>0.0.1-SNAPSHOT</version>
  <name/>
  <description/>
  <url/>
  <licenses>
    <license/>
  </licenses>
  <developers>
    <developer/>
  </developers>
  <scm>
    <connection/>
    <developerConnection/>
    <tag/>
    <url/>
  </scm>
  <properties>
    <java.version>21</java.version>
  </properties>
  <dependencies>
    <dependency>
      <groupId>org.springframework.boot</groupId>
      <artifactId>spring-boot-starter-webmvc</artifactId>
    </dependency>

    <dependency>
      <groupId>org.springframework.boot</groupId>
      <artifactId>spring-boot-devtools</artifactId>
      <scope>runtime</scope>
      <optional>true</optional>
    </dependency>
    <dependency>
```



```

        <groupId>org.springframework.boot</groupId>
        <artifactId>spring-boot-starter-webmvc-test</artifactId>
        <scope>test</scope>
    </dependency>
    <dependency>
        <groupId>org.springframework</groupId>
        <artifactId>spring-context</artifactId>
    </dependency>
</dependencies>

<build>
    <plugins>
        <plugin>
            <groupId>org.springframework.boot</groupId>
            <artifactId>spring-boot-maven-plugin</artifactId>
        </plugin>
    </plugins>
</build>
</project>

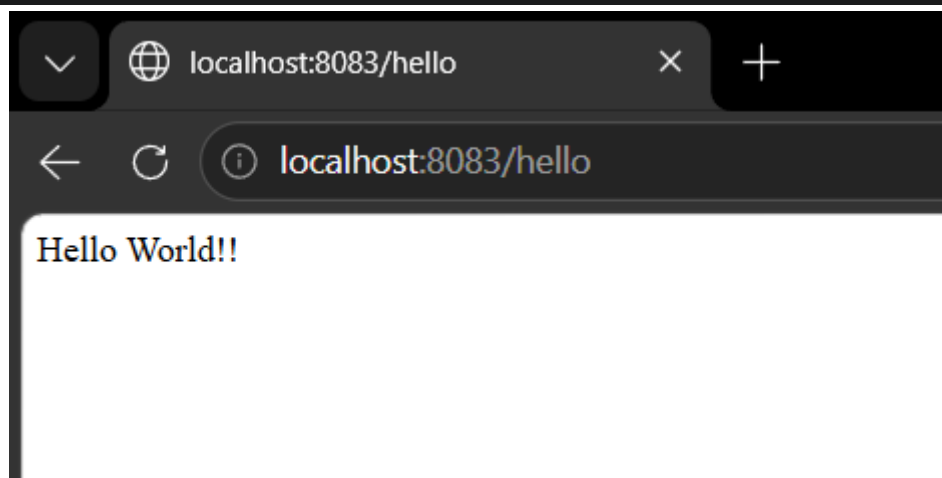
```

Output:

```

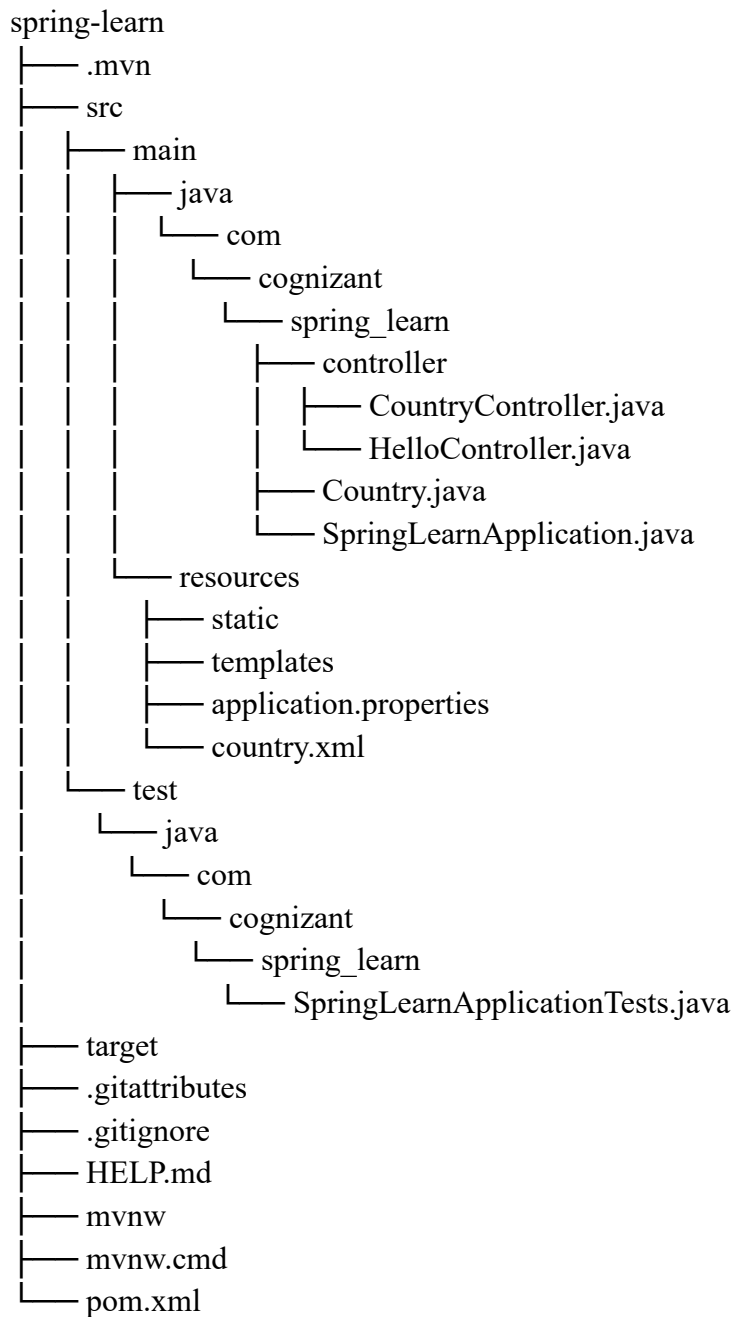
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
2026-07-26T18:26:48.747+05:30 INFO 12860 --- [ restartedMain] o.s.boot.tomcat.TomcatWebServer : Tomcat initialized with port 8083 (http)
2026-07-26T18:26:48.774+05:30 INFO 12860 --- [ restartedMain] o.apache.catalina.core.StandardService : Starting Service [Tomcat]
2026-07-26T18:26:48.775+05:30 INFO 12860 --- [ restartedMain] o.apache.catalina.core.StandardEngine : Starting Servlet engine: [Apache Tomcat/11.0.22]
2026-07-26T18:26:48.838+05:30 INFO 12860 --- [ restartedMain] b.w.c.s.WebApplicationContextInitializer : Root WebApplicationContext: initialization completed in 1392 ms
2026-07-26T18:26:49.408+05:30 INFO 12860 --- [ restartedMain] o.s.boot.tomcat.TomcatWebServer : Tomcat started on port 8083 (http) with context path '/'
2026-07-26T18:26:49.418+05:30 INFO 12860 --- [ restartedMain] c.c.spring_learn.SpringLearnApplication : Started SpringLearnApplication in 2.754 seconds (process running for 3.412)
2026-07-26T18:26:49.425+05:30 INFO 12860 --- [ restartedMain] c.c.spring_learn.SpringLearnApplication : Inside main
2026-07-26T18:26:49.425+05:30 INFO 12860 --- [ restartedMain] c.c.spring_learn.SpringLearnApplication : Start
2026-07-26T18:26:49.622+05:30 DEBUG 12860 --- [ restartedMain] com.cognizant.spring_learn.Country : Inside Country Constructor.
2026-07-26T18:26:49.623+05:30 DEBUG 12860 --- [ restartedMain] com.cognizant.spring_learn.Country : Inside setCode()
2026-07-26T18:26:49.623+05:30 DEBUG 12860 --- [ restartedMain] com.cognizant.spring_learn.Country : Inside setName()
2026-07-26T18:26:49.625+05:30 DEBUG 12860 --- [ restartedMain] c.c.spring_learn.SpringLearnApplication : Country : Country [code=IN, name=India]
2026-07-26T18:26:49.626+05:30 INFO 12860 --- [ restartedMain] c.c.spring_learn.SpringLearnApplication : End
2026-07-26T18:26:57.393+05:30 INFO 12860 --- [nio-8083-exec-1] o.a.c.c.C.[Tomcat].[localhost].[/] : Initializing Spring DispatcherServlet 'dispatcherServlet'
2026-07-26T18:26:57.393+05:30 INFO 12860 --- [nio-8083-exec-1] o.s.web.servlet.DispatcherServlet : Initializing Servlet 'dispatcherServlet'
2026-07-26T18:26:57.396+05:30 INFO 12860 --- [nio-8083-exec-1] o.s.web.servlet.DispatcherServlet : Completed initialization in 2 ms
2026-07-26T18:26:57.447+05:30 INFO 12860 --- [nio-8083-exec-1] c.c.s.controller.HelloController : Start
2026-07-26T18:26:57.448+05:30 INFO 12860 --- [nio-8083-exec-1] c.c.s.controller.HelloController : End
2026-07-26T18:28:06.304+05:30 INFO 12860 --- [nio-8083-exec-4] c.c.s.controller.HelloController : Start
2026-07-26T18:28:06.305+05:30 INFO 12860 --- [nio-8083-exec-4] c.c.s.controller.HelloController : End

```



4. Rest - Country Web Service

The Project Directory is as follows:



Codes:

CountryController.java

```
package com.cognizant.spring_learn.controller;
import org.slf4j.Logger;
import org.slf4j.LoggerFactory;
import org.springframework.context.ApplicationContext;
import org.springframework.context.support.ClassPathXmlApplicationContext;
import org.springframework.web.bind.annotation.RequestMapping;
import org.springframework.web.bind.annotation.RestController;
```

```
import com.cognizant.spring_learn.Country;
@RestController
public class CountryController {
    private static final Logger LOGGER =
        LoggerFactory.getLogger(CountryController.class);
    @RequestMapping("/country")
    public Country getCountryIndia() {
        LOGGER.info("Start");
        ApplicationContext context =
            new ClassPathXmlApplicationContext("country.xml");
        Country country =
            context.getBean("country", Country.class);
        LOGGER.info("End");
        ((ClassPathXmlApplicationContext) context).close();
        return country;
    }
}
```

Output:

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
2026-07-26T19:02:37.400+05:30 INFO 21920 --- [ restartedMain] o.s.boot.tomcat.TomcatWebServer : Tomcat initialized with port 8083 (http)
2026-07-26T19:02:37.427+05:30 INFO 21920 --- [ restartedMain] o.apache.catalina.core.StandardService : Starting service [Tomcat]
2026-07-26T19:02:37.428+05:30 INFO 21920 --- [ restartedMain] o.apache.catalina.core.StandardEngine : Starting Servlet engine: [Apache Tomcat/11.0.22]
2026-07-26T19:02:37.493+05:30 INFO 21920 --- [ restartedMain] b.w.c.s.WebApplicationContextInitializer : Root WebApplicationContext: initialization completed in 1318 ms
2026-07-26T19:02:38.027+05:30 INFO 21920 --- [ restartedMain] o.s.boot.tomcat.TomcatWebServer : Tomcat started on port 8083 (http) with context path '/'
2026-07-26T19:02:38.035+05:30 INFO 21920 --- [ restartedMain] c.c.spring_learn.SpringLearnApplication : Started SpringLearnApplication in 2.723 seconds (process running for 3.472)
2026-07-26T19:02:38.040+05:30 INFO 21920 --- [ restartedMain] c.c.spring_learn.SpringLearnApplication : Inside main
2026-07-26T19:02:38.041+05:30 INFO 21920 --- [ restartedMain] c.c.spring_learn.SpringLearnApplication : Start
2026-07-26T19:02:38.190+05:30 DEBUG 21920 --- [ restartedMain] com.cognizant.spring_learn.Country : Inside setCode()
2026-07-26T19:02:38.199+05:30 DEBUG 21920 --- [ restartedMain] com.cognizant.spring_learn.Country : Inside setName()
2026-07-26T19:02:38.200+05:30 DEBUG 21920 --- [ restartedMain] c.c.spring_learn.SpringLearnApplication : Country : Country [code=IN, name=India]
2026-07-26T19:02:38.201+05:30 INFO 21920 --- [ restartedMain] c.c.spring_learn.SpringLearnApplication : End
2026-07-26T19:02:52.627+05:30 INFO 21920 --- [nio-8083-exec-1] o.a.c.c.C.[Tomcat].[localhost].[/] : Initializing Spring DispatcherServlet 'dispatcherServlet'
2026-07-26T19:02:52.628+05:30 INFO 21920 --- [nio-8083-exec-1] o.s.web.servlet.DispatcherServlet : Initializing Servlet 'dispatcherServlet'
2026-07-26T19:02:52.630+05:30 INFO 21920 --- [nio-8083-exec-1] o.s.web.servlet.DispatcherServlet : Completed initialization in 1 ms
2026-07-26T19:02:52.693+05:30 INFO 21920 --- [nio-8083-exec-1] c.c.s.controller.CountryController : Start
2026-07-26T19:02:52.726+05:30 DEBUG 21920 --- [nio-8083-exec-1] com.cognizant.spring_learn.Country : Inside Country Constructor.
2026-07-26T19:02:52.727+05:30 DEBUG 21920 --- [nio-8083-exec-1] com.cognizant.spring_learn.Country : Inside setCode()
2026-07-26T19:02:52.728+05:30 DEBUG 21920 --- [nio-8083-exec-1] com.cognizant.spring_learn.Country : Inside setName()
2026-07-26T19:02:52.728+05:30 INFO 21920 --- [nio-8083-exec-1] c.c.s.controller.CountryController : End
2026-07-26T19:02:52.866+05:30 DEBUG 21920 --- [nio-8083-exec-1] com.cognizant.spring_learn.Country : Inside getCode()
2026-07-26T19:02:52.869+05:30 DEBUG 21920 --- [nio-8083-exec-1] com.cognizant.spring_learn.Country : Inside getName()
```

localhost:8083/country

localhost:8083/country

Pretty-print ☒

```
{
  "code": "IN",
  "name": "India"
}
```

5. REST – Get country based on country code

The full Project Directory is as follows:

```
spring-learn
├── .mvn
├── src
│   ├── main
│   │   ├── java
│   │   │   ├── com
│   │   │   │   ├── cognizant
│   │   │   │   │   ├── spring_learn
│   │   │   │   │   │   ├── controller
│   │   │   │   │   │   │   ├── CountryController.java
│   │   │   │   │   │   │   ├── HelloController.java
│   │   │   │   │   │   ├── service
│   │   │   │   │   │   │   ├── CountryService.java
│   │   │   │   │   │   ├── Country.java
│   │   │   │   │   │   └── SpringLearnApplication.java
│   │   │   │   │   └── resources
│   │   │   │   │       ├── static
│   │   │   │   │       ├── templates
│   │   │   │   │       ├── application.properties
│   │   │   │   │       └── country.xml
│   │   └── test
│   │       ├── java
│   │       │   ├── com
│   │       │   │   ├── cognizant
│   │       │   │   │   ├── spring_learn
│   │       │   │   │   └── SpringLearnApplicationTests.java
│   └── target
├── .gitattributes
├── .gitignore
├── HELP.md
├── mvnw
├── mvnw.cmd
└── pom.xml
```

Codes:

CountryController.java

```
package com.cognizant.spring_learn.controller;
import org.slf4j.Logger;
import org.slf4j.LoggerFactory;
import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.web.bind.annotation.GetMapping;
```

```

import org.springframework.web.bind.annotation.PathVariable;
import org.springframework.web.bind.annotation.RestController;
import com.cognizant.spring_learn.Country;
import com.cognizant.spring_learn.service.CountryService;
@RestController
public class CountryController {
    private static final Logger LOGGER =
        LoggerFactory.getLogger(CountryController.class);
    @Autowired
    private CountryService countryService;
    @GetMapping("/countries/{code}")
    public Country getCountry(@PathVariable String code) {
        LOGGER.info("Start");
        Country country = countryService.getCountry(code);
        LOGGER.info("End");
        return country;
    }
}

```

CountryService.java

```

package com.cognizant.spring_learn.service;
import java.util.List;
import org.springframework.context.ApplicationContext;
import org.springframework.context.support.ClassPathXmlApplicationContext;
import org.springframework.stereotype.Service;
import com.cognizant.spring_learn.Country;
@Service
public class CountryService {
    @SuppressWarnings("unchecked")
    public Country getCountry(String code) {
        ApplicationContext context =
            new ClassPathXmlApplicationContext("country.xml");
        List<Country> countries =
            (List<Country>) context.getBean("countryList");
        for (Country country : countries) {
            if (country.getCode().equalsIgnoreCase(code)) {
                ((ClassPathXmlApplicationContext) context).close();
                return country;
            }
        }
        ((ClassPathXmlApplicationContext) context).close();
        return null;
    }
}

```

country.xml

```

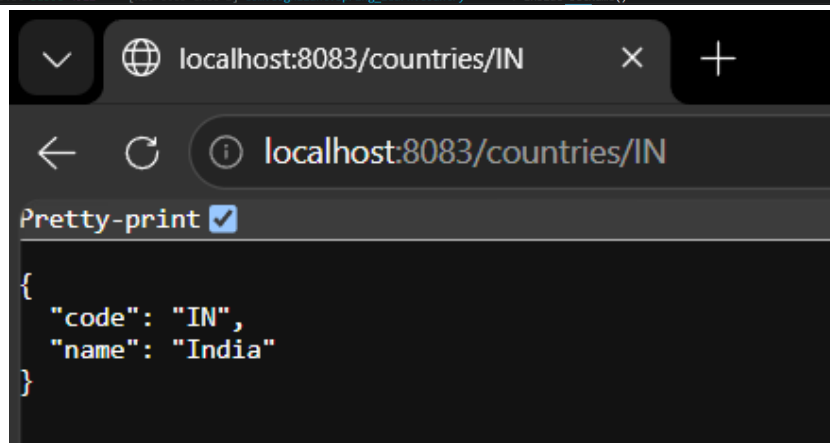
<?xml version="1.0" encoding="UTF-8"?>
<beans xmlns="http://www.springframework.org/schema/beans"
    xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"

```

```
xsi:schemaLocation="
http://www.springframework.org/schema/beans
https://www.springframework.org/schema/beans/spring-beans.xsd">
<bean id="countryList" class="java.util.ArrayList">
  <constructor-arg>
    <list>
      <bean class="com.cognizant.spring_learn.Country">
        <property name="code" value="US"/>
        <property name="name" value="United States"/>
      </bean>
      <bean class="com.cognizant.spring_learn.Country">
        <property name="code" value="DE"/>
        <property name="name" value="Germany"/>
      </bean>
      <bean class="com.cognizant.spring_learn.Country">
        <property name="code" value="IN"/>
        <property name="name" value="India"/>
      </bean>
      <bean class="com.cognizant.spring_learn.Country">
        <property name="code" value="JP"/>
        <property name="name" value="Japan"/>
      </bean>
    </list>
  </constructor-arg>
</bean>
</beans>
```

Outputs:

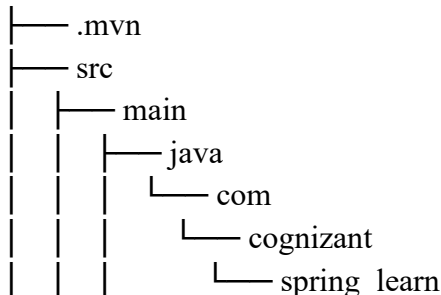
```
PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS
2026-07-26T19:27:33.569+05:30 INFO 4912 --- [ restartedMain] o.s.boot.tomcat.TomcatWebServer : Tomcat initialized with port 8083 (http)
2026-07-26T19:27:33.595+05:30 INFO 4912 --- [ restartedMain] o.apache.catalina.core.StandardService : Starting service [Tomcat]
2026-07-26T19:27:33.596+05:30 INFO 4912 --- [ restartedMain] o.apache.catalina.core.StandardEngine : Starting Servlet engine: [Apache Tomcat/11.0.22]
2026-07-26T19:27:33.660+05:30 INFO 4912 --- [ restartedMain] b.w.c.s.WebApplicationContextInitializer : Root WebApplicationContext: initialization completed in 1174 ms
2026-07-26T19:27:34.160+05:30 INFO 4912 --- [ restartedMain] o.s.boot.tomcat.TomcatWebServer : Tomcat started on port 8083 (http) with context path '/'
2026-07-26T19:27:34.175+05:30 INFO 4912 --- [ restartedMain] c.c.spring_learn.SpringLearnApplication : Started SpringLearnApplication in 2.452 seconds (process running for 3.169)
2026-07-26T19:27:34.175+05:30 INFO 4912 --- [ restartedMain] c.c.spring_learn.SpringLearnApplication : Inside main
2026-07-26T19:27:34.354+05:30 DEBUG 4912 --- [ restartedMain] com.cognizant.spring_learn.Country : Start
2026-07-26T19:27:34.355+05:30 DEBUG 4912 --- [ restartedMain] com.cognizant.spring_learn.Country : Inside Country Constructor.
2026-07-26T19:27:34.356+05:30 DEBUG 4912 --- [ restartedMain] com.cognizant.spring_learn.Country : Inside setCode()
2026-07-26T19:27:34.356+05:30 DEBUG 4912 --- [ restartedMain] com.cognizant.spring_learn.Country : Inside setName()
2026-07-26T19:27:34.356+05:30 DEBUG 4912 --- [ restartedMain] com.cognizant.spring_learn.Country : Inside Country Constructor.
2026-07-26T19:27:34.357+05:30 DEBUG 4912 --- [ restartedMain] com.cognizant.spring_learn.Country : Inside setCode()
2026-07-26T19:27:34.357+05:30 DEBUG 4912 --- [ restartedMain] com.cognizant.spring_learn.Country : Inside setName()
2026-07-26T19:27:34.357+05:30 DEBUG 4912 --- [ restartedMain] com.cognizant.spring_learn.Country : Inside Country Constructor.
2026-07-26T19:27:34.358+05:30 DEBUG 4912 --- [ restartedMain] com.cognizant.spring_learn.Country : Inside setCode()
2026-07-26T19:27:34.358+05:30 DEBUG 4912 --- [ restartedMain] com.cognizant.spring_learn.Country : Inside setName()
2026-07-26T19:27:34.358+05:30 DEBUG 4912 --- [ restartedMain] com.cognizant.spring_learn.Country : Inside Country Constructor.
2026-07-26T19:27:34.359+05:30 DEBUG 4912 --- [ restartedMain] com.cognizant.spring_learn.Country : Inside setCode()
2026-07-26T19:27:34.359+05:30 DEBUG 4912 --- [ restartedMain] com.cognizant.spring_learn.Country : Inside setName()
Exception in thread "restartedMain" java.lang.reflect.InvocationTargetException
    at java.base/jdk.internal.reflect.DirectMethodHandleAccessor.invoke(DirectMethodHandleAccessor.java:118)
    at java.base/java.lang.reflect.Method.invoke(Method.java:568)
    at org.springframework.boot.devtools.restart.RestartLauncher.run(RestartLauncher.java:52)
Caused by: org.springframework.beans.factory.NoSuchBeanDefinitionException: No bean named 'country' available
    at org.springframework.beans.factory.support.DefaultListableBeanFactory.getBeanDefinition(DefaultListableBeanFactory.java:971)
    at org.springframework.beans.factory.support.AbstractBeanFactory.getMergedLocalBeanDefinition(AbstractBeanFactory.java:1369)
    at org.springframework.beans.factory.support.AbstractBeanFactory.doGetBean(AbstractBeanFactory.java:296)
    at org.springframework.beans.factory.support.AbstractBeanFactory.getBean(AbstractBeanFactory.java:201)
    at org.springframework.context.support.AbstractApplicationContext.getBean(AbstractApplicationContext.java:1305)
    at com.cognizant.spring_learn.SpringLearnApplication.displayCountry(SpringLearnApplication.java:33)
    at com.cognizant.spring_learn.SpringLearnApplication.main(SpringLearnApplication.java:22)
    at java.base/jdk.internal.reflect.DirectMethodHandleAccessor.invoke(DirectMethodHandleAccessor.java:103)
    ... 2 more
2026-07-26T19:27:42.890+05:30 INFO 4912 --- [nio-8083-exec-1] o.a.c.c.C.[Tomcat].[localhost].[/] : Initializing Spring DispatcherServlet 'dispatcherServlet'
2026-07-26T19:27:42.890+05:30 INFO 4912 --- [nio-8083-exec-1] o.s.web.servlet.DispatcherServlet : Initializing Servlet 'dispatcherServlet'
2026-07-26T19:27:42.902+05:30 INFO 4912 --- [nio-8083-exec-1] o.s.web.servlet.DispatcherServlet : Completed initialization in 11 ms
2026-07-26T19:27:42.974+05:30 INFO 4912 --- [nio-8083-exec-1] c.c.s.controller.CountryController : Start
2026-07-26T19:27:43.007+05:30 DEBUG 4912 --- [nio-8083-exec-1] com.cognizant.spring_learn.Country : Inside Country Constructor.
2026-07-26T19:27:43.007+05:30 DEBUG 4912 --- [nio-8083-exec-1] com.cognizant.spring_learn.Country : Inside setCode()
2026-07-26T19:27:43.007+05:30 DEBUG 4912 --- [nio-8083-exec-1] com.cognizant.spring_learn.Country : Inside setName()
2026-07-26T19:27:43.007+05:30 DEBUG 4912 --- [nio-8083-exec-1] com.cognizant.spring_learn.Country : Inside Country Constructor.
2026-07-26T19:27:43.010+05:30 DEBUG 4912 --- [nio-8083-exec-1] com.cognizant.spring_learn.Country : Inside setCode()
2026-07-26T19:27:43.010+05:30 DEBUG 4912 --- [nio-8083-exec-1] com.cognizant.spring_learn.Country : Inside setName()
2026-07-26T19:27:43.010+05:30 DEBUG 4912 --- [nio-8083-exec-1] com.cognizant.spring_learn.Country : Inside Country Constructor.
2026-07-26T19:27:43.010+05:30 DEBUG 4912 --- [nio-8083-exec-1] com.cognizant.spring_learn.Country : Inside setCode()
2026-07-26T19:27:43.010+05:30 DEBUG 4912 --- [nio-8083-exec-1] com.cognizant.spring_learn.Country : Inside setName()
```

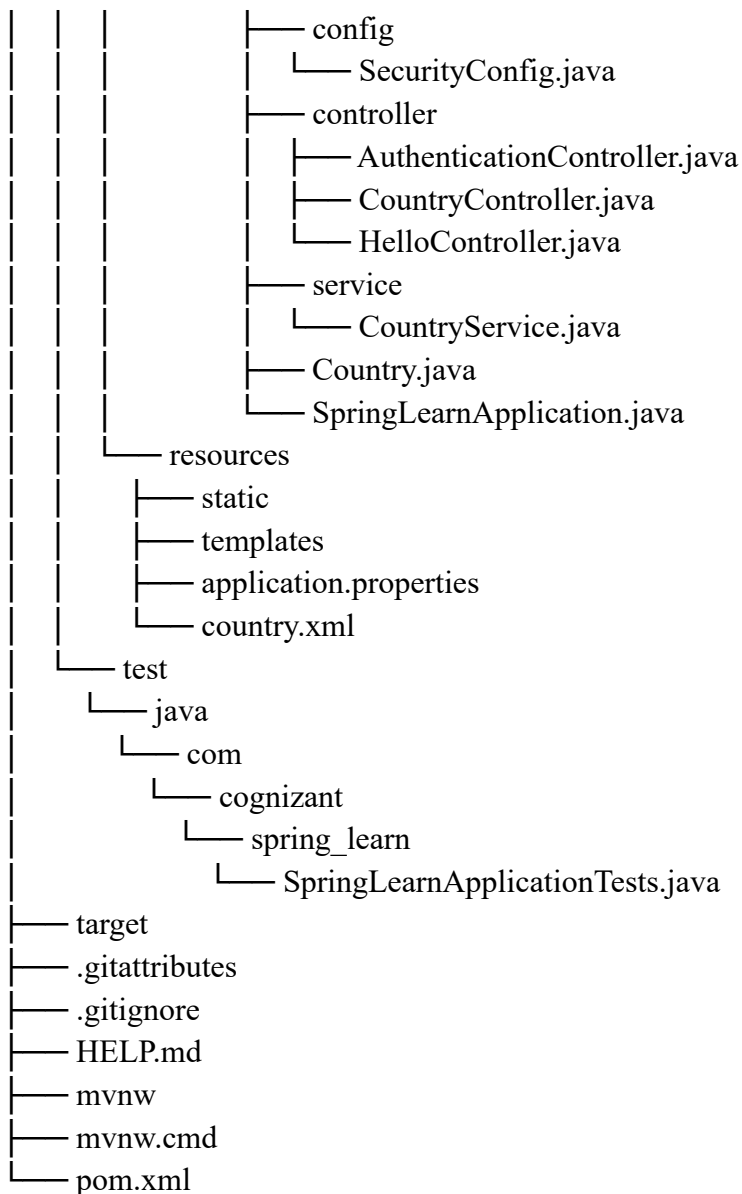


6. Create authentication service that returns JWT

The Project Directory is as follows:

spring-learn





Codes:

SecurityConfig.java

```
package com.cognizant.spring_learn.config;
import org.springframework.context.annotation.Bean;
import org.springframework.context.annotation.Configuration;
import org.springframework.security.config.Customizer;
import
org.springframework.security.config.annotation.web.builders.HttpSecurity;
import org.springframework.security.core.userdetails.User;
import org.springframework.security.core.userdetails.UserDetails;
import org.springframework.security.provisioning.InMemoryUserDetailsManager;
import org.springframework.security.web.SecurityFilterChain;
@Configuration
public class SecurityConfig {
    @Bean
    public InMemoryUserDetailsManager userDetailsService() {
```



```

        UserDetails user = User.withDefaultPasswordEncoder()
            .username("user")
            .password("pwd")
            .roles("USER")
            .build();
        UserDetails admin = User.withDefaultPasswordEncoder()
            .username("admin")
            .password("pwd")
            .roles("ADMIN")
            .build();
        return new InMemoryUserDetailsManager(user, admin);
    }
    @Bean
    public SecurityFilterChain securityFilterChain(HttpSecurity http)
        throws Exception {
        http
            .csrf(csrf -> csrf.disable())
            .authorizeHttpRequests(auth -> auth
                .requestMatchers("/authenticate")
                .hasAnyRole("USER", "ADMIN")
                .requestMatchers("/country")
                .hasRole("USER")
                .requestMatchers("/countries/**")
                .hasRole("USER")
                .requestMatchers("/hello")
                .permitAll()
                .anyRequest()
                .authenticated()
            )
            .httpBasic(Customizer.withDefaults());
        return http.build();
    }
}

```

AuthenticationController.java

```

package com.cognizant.spring_learn.controller;
import java.util.Base64;
import java.util.Date;
import java.util.HashMap;
import java.util.Map;
import org.slf4j.Logger;
import org.slf4j.LoggerFactory;
import org.springframework.web.bind.annotation.GetMapping;
import org.springframework.web.bind.annotation.RequestHeader;
import org.springframework.web.bind.annotation.RestController;
import io.jsonwebtoken.JwtBuilder;
import io.jsonwebtoken.Jwts;
import io.jsonwebtoken.SignatureAlgorithm;
@RestController

```

```

public class AuthenticationController {
    private static final Logger LOGGER =
        LoggerFactory.getLogger(AuthenticationController.class);
    @GetMapping("/authenticate")
    public Map<String, String> authenticate(
        @RequestHeader("Authorization") String authHeader) {
        LOGGER.info("Start");
        LOGGER.debug("Authorization Header : {}", authHeader);
        String user = getUser(authHeader);
        String token = generateJwt(user);
        Map<String, String> map = new HashMap<>();
        map.put("token", token);
        LOGGER.info("End");
        return map;
    }
    private String getUser(String authHeader) {
        LOGGER.debug("Inside getUser()");
        String encodedCredentials =
            authHeader.substring("Basic ".length());
        byte[] decodedBytes =
            Base64.getDecoder().decode(encodedCredentials);
        String credentials =
            new String(decodedBytes);
        LOGGER.debug("Decoded Credentials : {}", credentials);
        String user =
            credentials.split(":")[0];
        LOGGER.debug("User : {}", user);
        return user;
    }
    private String generateJwt(String user) {
        LOGGER.debug("Generating JWT");
        JwtBuilder builder = Jwts.builder();
        builder.setSubject(user);
        builder.setIssuedAt(new Date());
        builder.setExpiration(
            new Date(new Date().getTime() + 1200000));
        builder.signWith(
            SignatureAlgorithm.HS256,
            "secretkey");
        String token = builder.compact();
        LOGGER.debug("Token Generated");
        return token;
    }
}

```

CountryController.java

```

package com.cognizant.spring_learn.controller;
import org.slf4j.Logger;
import org.slf4j.LoggerFactory;

```

```

import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.web.bind.annotation.GetMapping;
import org.springframework.web.bind.annotation.PathVariable;
import org.springframework.web.bind.annotation.RequestMapping;
import org.springframework.web.bind.annotation.RestController;
import com.cognizant.spring_learn.Country;
import com.cognizant.spring_learn.service.CountryService;
@RestController
public class CountryController {
    private static final Logger LOGGER =
        LoggerFactory.getLogger(CountryController.class);
    @Autowired
    private CountryService countryService;
    // Previous assignment
    @RequestMapping("/country")
    public Country getCountryIndia() {
        LOGGER.info("Start");
        Country country = countryService.getCountry("IN");
        LOGGER.info("End");
        return country;
    }
    // Current assignment
    @GetMapping("/countries/{code}")
    public Country getCountry(@PathVariable String code) {
        LOGGER.info("Start");
        Country country = countryService.getCountry(code);
        LOGGER.info("End");
        return country;
    }
}

```

HelloController.java

```

package com.cognizant.spring_learn.controller;
import org.slf4j.Logger;
import org.slf4j.LoggerFactory;
import org.springframework.web.bind.annotation.GetMapping;
import org.springframework.web.bind.annotation.RestController;
@RestController
public class HelloController {
    private static final Logger LOGGER =
        LoggerFactory.getLogger(HelloController.class);
    @GetMapping("/hello")
    public String sayHello() {
        LOGGER.info("Start");
        LOGGER.info("End");
        return "Hello World!!";
    }
}

```

CountryService.java

```
package com.cognizant.spring_learn.service;
import java.util.List;
import org.springframework.context.ApplicationContext;
import org.springframework.context.support.ClassPathXmlApplicationContext;
import org.springframework.stereotype.Service;
import com.cognizant.spring_learn.Country;
@Service
public class CountryService {
    @SuppressWarnings("unchecked")
    public Country getCountry(String code) {
        ApplicationContext context =
            new ClassPathXmlApplicationContext("country.xml");
        List<Country> countries =
            (List<Country>) context.getBean("countryList");
        for (Country country : countries) {
            if (country.getCode().equalsIgnoreCase(code)) {
                ((ClassPathXmlApplicationContext) context).close();
                return country;
            }
        }
        ((ClassPathXmlApplicationContext) context).close();
        return null;
    }
}
```

Country.java

```
package com.cognizant.spring_learn;
import org.slf4j.Logger;
import org.slf4j.LoggerFactory;
public class Country {
    private static final Logger LOGGER =
        LoggerFactory.getLogger(Country.class);
    private String code;
    private String name;
    public Country() {
        LOGGER.debug("Inside Country Constructor.");
    }
    public String getCode() {
        LOGGER.debug("Inside getCode()");
        return code;
    }
    public void setCode(String code) {
        LOGGER.debug("Inside setCode()");
        this.code = code;
    }
    public String getName() {
        LOGGER.debug("Inside getName()");
        return name;
    }
}
```

```

    }
    public void setName(String name) {
        LOGGER.debug("Inside setName()");
        this.name = name;
    }
    @Override
    public String toString() {
        return "Country [code=" + code + ", name=" + name + "]";
    }
}

```

SpringLearnApplication.java

```

package com.cognizant.spring_learn;
import org.slf4j.Logger;
import org.slf4j.LoggerFactory;
import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;
import org.springframework.context.ApplicationContext;
import org.springframework.context.support.ClassPathXmlApplicationContext;
@SpringBootApplication
public class SpringLearnApplication {
    private static final Logger LOGGER =
        LoggerFactory.getLogger(SpringLearnApplication.class);
    public static void main(String[] args) {
        SpringApplication.run(SpringLearnApplication.class, args);
        LOGGER.info("Inside main");
        displayCountry();
    }
    public static void displayCountry() {
        LOGGER.info("Start");
        ApplicationContext context =
            new ClassPathXmlApplicationContext("country.xml");
        Country country =
            context.getBean("country", Country.class);
        LOGGER.debug("Country : {}", country);
        LOGGER.info("End");
        ((ClassPathXmlApplicationContext) context).close();
    }
}

```

application.properties

```

logging.level.org.springframework=INFO
logging.level.com.cognizant=DEBUG
server.port=8090

```

country.xml

```

<?xml version="1.0" encoding="UTF-8"?>
<beans xmlns="http://www.springframework.org/schema/beans"
    xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
    xsi:schemaLocation="

```

```

    http://www.springframework.org/schema/beans
    https://www.springframework.org/schema/beans/spring-beans.xsd">
    <bean id="countryList" class="java.util.ArrayList">
        <constructor-arg>
            <list>
                <bean class="com.cognizant.spring_learn.Country">
                    <property name="code" value="US"/>
                    <property name="name" value="United States"/>
                </bean>
                <bean class="com.cognizant.spring_learn.Country">
                    <property name="code" value="DE"/>
                    <property name="name" value="Germany"/>
                </bean>
                <bean class="com.cognizant.spring_learn.Country">
                    <property name="code" value="IN"/>
                    <property name="name" value="India"/>
                </bean>
                <bean class="com.cognizant.spring_learn.Country">
                    <property name="code" value="JP"/>
                    <property name="name" value="Japan"/>
                </bean>
            </list>
        </constructor-arg>
    </bean>
</beans>

```

SpringLearnApplicationTests.java

```

package com.cognizant.spring_learn;
import org.junit.jupiter.api.Test;
import org.springframework.boot.test.context.SpringBootTest;
@SpringBootTest
class SpringLearnApplicationTests {
    @Test
    void contextLoads() {
    }
}

```

Outputs:

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
2026-07-26T20:22:05.236+05:30 INFO 27344 --- [ restartedMain] o.s.boot.tomcat.TomcatWebServer : Tomcat initialized with port 8090 (http)
2026-07-26T20:22:05.274+05:30 INFO 27344 --- [ restartedMain] o.apache.catalina.core.StandardService : Starting service [Tomcat]
2026-07-26T20:22:05.276+05:30 INFO 27344 --- [ restartedMain] o.apache.catalina.core.StandardEngine : Starting Servlet engine: [Apache Tomcat/11.0.22]
2026-07-26T20:22:05.309+05:30 INFO 27344 --- [ restartedMain] b.w.c.s.WebApplicationContextInitializer : Root WebApplicationContext: initialization completed in 1781 ms
2026-07-26T20:22:05.352+05:30 WARN 27344 --- [ restartedMain] o.s.security.core.userdetails.User : UserWithDefaultPasswordEncoder() is considered unsafe for production and is only intended for sample applications.
2026-07-26T20:22:05.352+05:30 WARN 27344 --- [ restartedMain] o.s.security.core.userdetails.User : UserWithDefaultPasswordEncoder() is considered unsafe for production and is only intended for sample applications.
2026-07-26T20:22:05.381+05:30 INFO 27344 --- [ restartedMain] i.InitializationManagerConfigurer : Global InitializationManager configured with UserDetailsServiceImpl bean with name userDetailsService
2026-07-26T20:22:05.839+05:30 INFO 27344 --- [ restartedMain] o.s.boot.tomcat.TomcatWebServer : Tomcat started on port 8090 (http) with context path '/'
2026-07-26T20:22:06.849+05:30 INFO 27344 --- [ restartedMain] c.c.spring_learn.SpringLearnApplication : Started SpringLearnApplication in 4.159 seconds (process running for 4.966)
2026-07-26T20:22:06.858+05:30 INFO 27344 --- [ restartedMain] c.c.spring_learn.SpringLearnApplication : Inside main
2026-07-26T20:22:06.869+05:30 INFO 27344 --- [ restartedMain] c.c.spring_learn.SpringLearnApplication : Start
2026-07-26T20:22:07.121+05:30 DEBUG 27344 --- [ restartedMain] com.cognizant.spring_learn.Country : Inside Country Constructor.
2026-07-26T20:22:07.124+05:30 DEBUG 27344 --- [ restartedMain] com.cognizant.spring_learn.Country : Inside setCode()
2026-07-26T20:22:07.124+05:30 DEBUG 27344 --- [ restartedMain] com.cognizant.spring_learn.Country : Inside setName()
2026-07-26T20:22:07.124+05:30 DEBUG 27344 --- [ restartedMain] com.cognizant.spring_learn.Country : Inside Country Constructor.
2026-07-26T20:22:07.124+05:30 DEBUG 27344 --- [ restartedMain] com.cognizant.spring_learn.Country : Inside setCode()
2026-07-26T20:22:07.124+05:30 DEBUG 27344 --- [ restartedMain] com.cognizant.spring_learn.Country : Inside setName()
2026-07-26T20:22:07.124+05:30 DEBUG 27344 --- [ restartedMain] com.cognizant.spring_learn.Country : Inside Country Constructor.
2026-07-26T20:22:07.124+05:30 DEBUG 27344 --- [ restartedMain] com.cognizant.spring_learn.Country : Inside setCode()
2026-07-26T20:22:07.124+05:30 DEBUG 27344 --- [ restartedMain] com.cognizant.spring_learn.Country : Inside setName()
2026-07-26T20:22:07.128+05:30 DEBUG 27344 --- [ restartedMain] com.cognizant.spring_learn.Country : Inside setCode()
2026-07-26T20:22:07.128+05:30 DEBUG 27344 --- [ restartedMain] com.cognizant.spring_learn.Country : Inside setName()
2026-07-26T20:22:07.128+05:30 DEBUG 27344 --- [ restartedMain] com.cognizant.spring_learn.Country : Inside setCode()
2026-07-26T20:22:07.128+05:30 DEBUG 27344 --- [ restartedMain] com.cognizant.spring_learn.Country : Inside setName()
Exception in thread " restartedMain" java.lang.reflect.InvocationTargetException
at java.base/jdk.internal.reflect.DirectMethodHandleAccessor.invoke(DirectMethodHandleAccessor.java:118)
at java.base/java.lang.reflect.Method.invoke(Method.java:580)
at org.springframework.boot.devtools.restart.RestartLauncher.run(RestartLauncher.java:52)
Caused by: org.springframework.beans.factory.NoSuchBeanDefinitionException: no bean named 'country' available
at org.springframework.beans.factory.support.DefaultListableBeanFactory.getBeanDefinition(DefaultListableBeanFactory.java:971)
at org.springframework.beans.factory.support.AbstractBeanFactory.getMergedLocalBeanDefinition(AbstractBeanFactory.java:1369)
at org.springframework.beans.factory.support.AbstractBeanFactory.doGetBean(AbstractBeanFactory.java:296)
at org.springframework.beans.factory.support.AbstractBeanFactory.getBean(AbstractBeanFactory.java:201)
at org.springframework.context.support.AbstractApplicationContext.getBean(AbstractApplicationContext.java:1305)
at com.cognizant.spring_learn.SpringLearnApplication.displayCountry(SpringLearnApplication.java:33)
at com.cognizant.spring_learn.SpringLearnApplication.main(SpringLearnApplication.java:22)
at java.base/jdk.internal.reflect.DirectMethodHandleAccessor.invoke(DirectMethodHandleAccessor.java:103)
... 2 more
2026-07-26T20:22:08.699+05:30 INFO 27344 --- [nio-8090-exec-1] o.a.c.c. [Tomcat] [localhost/] : Initializing Spring DispatcherServlet 'dispatcherServlet'
2026-07-26T20:22:08.699+05:30 INFO 27344 --- [nio-8090-exec-1] o.s.web.servlet.DispatcherServlet : Initializing Servlet 'dispatcherServlet'
2026-07-26T20:22:08.702+05:30 INFO 27344 --- [nio-8090-exec-1] o.s.web.servlet.DispatcherServlet : Completed initialization in 2 ms
2026-07-26T20:22:08.777+05:30 INFO 27344 --- [nio-8090-exec-1] c.c.s.controller.HelloController : Start
2026-07-26T20:22:08.777+05:30 INFO 27344 --- [nio-8090-exec-1] c.c.s.controller.HelloController : End
2026-07-26T20:22:08.777+05:30 INFO 27344 --- [nio-8090-exec-4] c.c.s.controller.HelloController : Start
2026-07-26T20:22:08.777+05:30 INFO 27344 --- [nio-8090-exec-4] com.cognizant.spring_learn.Country : Inside Country Constructor.
2026-07-26T20:22:08.777+05:30 INFO 27344 --- [nio-8090-exec-4] com.cognizant.spring_learn.Country : Inside setCode()
2026-07-26T20:22:08.777+05:30 INFO 27344 --- [nio-8090-exec-4] com.cognizant.spring_learn.Country : Inside setName()
2026-07-26T20:22:08.777+05:30 INFO 27344 --- [nio-8090-exec-4] com.cognizant.spring_learn.Country : Inside Country Constructor.
2026-07-26T20:22:08.777+05:30 INFO 27344 --- [nio-8090-exec-4] com.cognizant.spring_learn.Country : Inside setCode()
2026-07-26T20:22:08.777+05:30 INFO 27344 --- [nio-8090-exec-4] com.cognizant.spring_learn.Country : Inside setName()
2026-07-26T20:22:08.777+05:30 INFO 27344 --- [nio-8090-exec-4] com.cognizant.spring_learn.Country : Inside Country Constructor.
2026-07-26T20:22:08.777+05:30 INFO 27344 --- [nio-8090-exec-4] com.cognizant.spring_learn.Country : Inside setCode()
2026-07-26T20:22:08.777+05:30 INFO 27344 --- [nio-8090-exec-4] com.cognizant.spring_learn.Country : Inside setName()
2026-07-26T20:22:08.777+05:30 INFO 27344 --- [nio-8090-exec-4] c.c.s.controller.HelloController : End
2026-07-26T20:22:08.777+05:30 INFO 27344 --- [nio-8090-exec-4] com.cognizant.spring_learn.Country : Inside setCode()
2026-07-26T20:22:08.777+05:30 INFO 27344 --- [nio-8090-exec-4] com.cognizant.spring_learn.Country : Inside setName()
2026-07-26T20:22:08.777+05:30 INFO 27344 --- [nio-8090-exec-4] com.cognizant.spring_learn.Country : Inside Country Constructor.
2026-07-26T20:22:08.777+05:30 INFO 27344 --- [nio-8090-exec-4] com.cognizant.spring_learn.Country : Inside setCode()
2026-07-26T20:22:08.777+05:30 INFO 27344 --- [nio-8090-exec-4] com.cognizant.spring_learn.Country : Inside setName()
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2026-07-26T20:22:08.777+05:30 INFO 27344 --- [nio-8090-exec-4] com.cognizant.spring_learn.Country : Inside setName()
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2026-07-26T20:22:08.777+05:30 INFO 27344 --- [nio-8090-exec-4] com.cognizant.spring_learn.Country : Inside setCode()
2026-07-26T20:22:08.777+05:30 INFO 27344 --- [nio-8090-exec-4] com.cognizant.spring_learn.Country : Inside setName()
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2026-07-26T20:22:08.777+05:30 INFO 27344 --- [nio-8090-exec-4] com.cognizant.spring_learn.Country : Inside setCode()
2026-07-26T20:22:08.777+05:30 INFO 27344 --- [nio-8090-exec-4] com.cognizant.spring_learn.Country : Inside setName()
2026-07-26T20:22:08.777+05:30 INFO 27344 --- [nio-8090-exec-4] com.cognizant.spring_learn.Country : Inside Country Constructor.
2026-07-26T20:22:08.777+05:30 INFO 27344 --- [nio-8090-exec-4] com.cognizant.spring_learn.Country : Inside setCode()
2026-07-26T20:22:08.777+05:30 INFO 27344 --- [nio-8090-exec-4] com.cognizant.spring_learn.Country : Inside setName()
2026-07-26T20:22:08.777+05:30 INFO 27344 --- [nio-8090-exec-4] com.cognizant.spring_learn.Country : Inside Country Constructor.
2026-07-26T20:22:08.777+05:30 INFO 27344 --- [nio-8090-exec-4] com.cognizant.spring_learn.Country : Inside setCode()
2026-07-26T20:22:08.777+05:30 INFO 27344 --- [nio-8090-exec-4] com.cognizant.spring_learn.Country : Inside setName()
2026-07-26T20:22:08.777+05:30 INFO 27344 --- [nio-8090-exec-4] com.cognizant.spring_learn.Country : Inside Country Constructor.
2026-07-26T20:22:08.777+05:30 INFO 27344 --- [nio-8090-exec-4] com.cognizant.spring_learn.Country : Inside setCode()
2026-07-26T20:22:08.777+05:30 INFO 273
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